Introduction to Psycholinguistics

Unit 3

Spoken Word Recognition

- Words are often treated as the basic building blocks of language.
- Even though, technically, a word isn't the smallest unit of meaning (that's the morpheme), in everyday thinking, we treat separate words as carrying separate meanings.
- Tyler & Frauenfelder (1987) divide the overall process of spoken word recognition into **five neat, manageable stages**.

They are **not necessarily temporally distinct** from one another, and they interact with other processes involved in understanding larger stretches of language, such as whole sentences.

Process of spoken word recognition



- the set of operations carried out on the raw incoming speech signal in order to organize it into linguistically useful units (e.g., phonemes, features, syllables
- establishing links between the (now pre-lexically-analyzed) input and the stored forms of words sitting in the mental lexicon.
- once contact is made, the contacted words become 'activated' essentially, they get excited about the fact that they have been contacted by matching input.
- actually getting hold of the information that is stored about a word in the mental lexicon: its meaning, its grammatical category (noun, verb, etc.), and other lexical properties.
- the final outcome: actually knowing which specific word it is that we have heard, out of all the candidates that were activated.

not strictly sequential boxes that speech passes through one at a time there is overlap and interaction between them

What Are Words?

- Cross-linguistic variation in what counts as a word:
- The concept of 'word' differs across languages.
- **Aymara is** a language spoken in the Chilean Andes. In Aymara, the root *uta* simply means **'house.'**
- But endings can be added onto that root to build up what would require an entire *sentence* in English:
  - *utawa* = **'it's a house'**
  - *utamanwa* = **'I see it's in your house'**
  - *utamanpachawa* = **'I see evidence to suggest that it's in your house'**
- In English, we'd call that a whole sentence made of several words; in Aymara, it's a single word built through extensive suffixation. So 'word' is not a language-neutral, self-evident unit.

Think

- Even if we all agree that words are 'things separated by spaces in printed text, are *cat* and *cats* two completely separate words, or two versions ('forms') of the same underlying word?
- Is *foot* a different word from *feet*, or the same word in a different form?
- Is *houseboat* a distinct word from both *house* and *boat*, or is it simply built out of them?

- **The minimal true unit of meaning is the morpheme**, not the word. A morpheme can include endings that can't stand alone but still carry meaning- for example, the plural -*s* or the past tense -*ed*. This matters because it shows that 'word' is a somewhat higher-level, socially/practically convenient unit, layered on top of more basic meaning-bearing pieces.

The Problem of Defining 'Knowing' a Word

- A key distinction between **passive vocabulary vs. active vocabulary.** Most speakers can **understand** far more words than they would ever actually **produce** in their own speech. Comprehension vocabulary > production vocabulary
- There's also a **modality mismatch**: there are words we recognize and understand when we see them in print, but that we've never actually encountered spoken aloud, and, occasionally, the reverse (words we know from speech but wouldn't recognize in writing).

  Reading vocabulary ≠ Spoken vocabulary

- How many words do people know? conservative estimate: **20,000 words** (Nation, 2006)

- Imagine someone has a mental lexicon of 20,000 words, and imagine they can search through that lexicon at a rate of **100 words per second**.
- If recognizing a word meant exhaustively comparing an input against *every single word we know, one at a time*, then determining that a nonsense string like ***splundle*** is **not** a real word of English would take about **3 minutes** of searching.
- But in reality, we reject *splundle* as a non-word almost instantly, nowhere close to 3 minutes.

- whatever the human word-recognition system is doing, **it is definitely not exhaustively matching the input against every word we know, one at a time.** It must be doing something far more efficient, narrowing down candidates rapidly rather than checking everything.

Pre-Lexical Analysis: Breaking Speech into Usable Units

- Pre-lexical analysis refers to the automatic, peripheral perceptual processes that analyze the raw spoken input into linguistically relevant units. Units, which are then used for word recognition and further comprehension.
- But, why not just use the whole word as the unit of analysis?

Challenges of using ‘word’ as unit of analysis

1. **The segmentation problem:**

In continuous spoken language, it's often extremely difficult to know exactly where one word ends and the next one begins — there are no built-in 'spaces' the way there are in written text. Example: the spoken phrase *'a coke and a Danish'* gets misheard as *'a coconut Danish.* **(Bond, 1999)**

2. **Waiting for the whole word would be too slow:**

If word-by-word analysis were used, a word could not be recognized until its *entire* speech pattern had been fully received.

The Advantages of Smaller, Sub-Word Units

- Analyzing speech pre-lexically into something smaller, like phonemes, has two major efficiency advantages:
- We can **begin** the recognition process on the basis of a much shorter stretch of speech than if we had to wait for an entire word.
- There are **far fewer distinct units** that the system needs to be able to recognize and tell apart.

The Phoneme as the Default Pre-Lexical Unit

- the phoneme is the *smallest* unit that, when changed, results in a change of meaning i.e., it signals a different word entirely. This is demonstrated using **minimal pairs**.
- **Definition of minimal pair:** a pair of words accepted as different (because they mean different things) that differ from one another in only one single sound.
- ***cat* vs. *cap*** : these mean different things and differ only in their **final** phoneme: /t/ vs. /p/.
- ***cat* vs. *cot*** : these differ in their **vowel** phoneme: /æ/ vs. /ɔ/ (in the author's variety of English).

Table 8.1 Consonant and vowel phonemes of New Zealand English, with example words containing these phonemes (as underlined).

consonants				vowels			
p	pin	ʃ	shore	ɪ	hid	eɪ	hay
b	bin	ʒ	leisure	ʊ	hood	əʊ	hoe
t	tin	h	ham	ɛ	hed	ɔɪ	boy
d	din	ʧ	chore	ɔ	hod	aɪ	high
k	kin	ʤ	judge	ʌ	bud	aʊ	how
g	gone	m	mow	æ	had	ɪə	here
f	fan	n	now	ə	again	ɛə	hair
v	van	ŋ	sing	i	heed	ʊə	tour
θ	thin	l	low	u	brood		
ð	than	r	row	ɜ	herd		
s	sue	w	woo	o	hoard		
z	zoo	j	you	a	hard		

A Challenge to the Phoneme's Special Status

- Marslen-Wilson & Warren (1994) raise a serious challenge: they argue that **phonemes may actually be linguistic constructs,** and that to the extent they exist as psychologically real entities in the mind at all, this might be a *consequence of having learned to read* (i.e., of making conscious connections between the sounds of spoken words and the shapes of written letters), rather than something inherent to speech processing itself.

- **Chinese** uses an **ideographic** writing system, where a single character represents a whole word/morpheme rather than representing individual sounds, there is very little representation of individual phonemes built into the writing system itself.
- **Experimental evidence (Morais, Cary, Alegria & Bertelson, 1979):** when speakers of such languages are asked to do tasks that require operating on phoneme-sized chunks of speech, for instance, a **phoneme monitoring task**, where participants must press a button the instant they hear a specific target sound such as /p/, they perform noticeably **worse** than literate speakers of alphabetic languages like English.

- This pattern suggests that conscious awareness of phonemes, and the ability to do phoneme-based tasks, is at least partly a *product of becoming literate in an alphabetic writing system*, not a universal prerequisite for spoken word recognition. Since non-literate speakers of non-alphabetic languages can nonetheless recognize and understand spoken words perfectly well, this is taken as evidence that phonemes are not strictly necessary for spoken word recognition; recognition can apparently proceed without an explicit phonemic level.

Smaller-Than-Phoneme Units: Phonetic Features

- Alternatives to the phoneme have been proposed at a *smaller* grain size too, most notably the **phonetic feature**.

1. Voicing: this is the feature that distinguishes /p/ from /b/ in English. There are many acoustic cues that signal this voicing contrast; one of them, discussed in the earlier speech-perception chapter, is **voice onset time (VOT)**.

- **2. Nasality:** nasality is the shared feature of the English sounds **/m, n, ŋ/**. During the production of these sounds, the soft palate (velum) at the back of the mouth is lowered, opening the passage between the mouth and nose and allowing air to flow out through the nose.
- English itself does **not** have contrasting oral vs. nasal *vowels* the way some other languages do; you can't make a new English word that differs from *cat* purely by nasalizing the vowel.
- **But,** English vowels **can become nasalised** as a *coarticulation* effect: their normally oral quality gets modified by allowing some air through the nose, for part or all of the vowel's duration, and this has a clearly noticeable effect on how the vowel sounds. This commonly happens when the vowel is immediately **followed by a nasal consonant.**

- **Example: the word 'soon':** the lowering of the velum (and resulting nasalization of the vowel) can actually begin **during the vowel itself**, before the /n/ has even been produced.
- **Experimental evidence- Gating experiments:** listeners can use this early nasalization cue to *anticipate* what the upcoming final consonant will be. Gating experiments (where increasingly long fragments of a recorded word are played and participants guess the word at each 'gate') show that a word like ***soon*** becomes identifiable *as* 'soon' - as opposed to *soup*, *suit*, or *sued* — **during the vowel portion**, i.e., *before* the final /n/ consonant has actually been heard (Warren & Marslen-Wilson, 1987, 1988).

Larger-Than-Phoneme Units

- Pre-lexical units *larger* than the phoneme have also been proposed, either as an alternative to, or a supplement to, phonemes. They are **diphones** and **syllables.**

Diphones

- diphones include the *transition* from one phoneme-sized segment to the next- meaning they naturally capture the variation in pronunciation caused by neighboring sounds (coarticulation), which a pure phoneme-by-phoneme analysis would lose.
- **example:** the /u/ sound in ***soon*** is likely to show nasalization because of the following nasal consonant. Similarly, the ending of that same /u/ sound differs in **quality** between ***soon/suit*** (where the following consonant is made with the tongue against the alveolar ridge- the bony structure just behind the top teeth) versus ***soup*** (where the following consonant involves a lip closure instead). So the *very same* vowel sound is subtly different across these three words purely because of what phoneme comes next, information a diphone-sized unit would directly encode, but a strict phoneme-sized unit would not.

Syllables (and Stressed Syllables specifically)

Cross-linguistic variation in syllable structure itself:

- **English** allows a wide range of syllable shapes: a syllable can be just a single vowel (*eye*), a vowel with one preceding or following consonant (*my*, *up*), or a vowel with multiple consonants both before and/or after it (*strengths*).
- **Māori**, by contrast, has a much more restricted syllable inventory: a Māori syllable **must** contain a vowel, which can be preceded by **one, and only one**, consonant, but **cannot** be followed by any consonant at all.
- **Implication for the model:** if the pre-lexical unit of analysis is the syllable, then speakers of different languages will literally be monitoring incoming speech for differently-sized and differently-structured chunks, because 'syllable' itself isn't a fixed, universal shape.

- **Cross-linguistic variation in stress placement:** Some languages have **fixed stress,** the stressed syllable's position is entirely predictable from the word's shape:
    - **Czech-** stress always falls on the **first** syllable.
    - **Polish-** stress always falls on the **second-to-last (penultimate)** syllable.
- Some languages have **no word stress at all.**
- **English** does have word stress, and it *is* predictable for any given word (dictionaries can mark it), but it is **not** fixed to the same position across all words:
    - **Worked example- a genuine minimal contrast:** the **noun** *import* has stress on the **first** syllable, while the **verb** *import* has stress on the **second** syllable.
- **Statistical regularity, despite the unpredictability of position:** corpus analyses show that a clear majority, roughly **90%,** of English content words have a strong (stressed) **first** syllable (Cutler & Carter, 1987). This statistical tendency becomes the empirical foundation for the segmentation strategy discussed next.

The Metrical Segmentation Strategy (MSS)

- **(Cutler & Norris, 1988):** because ~90% of English content words begin with a stressed syllable, the word-recognition system exploits this regularity by **initiating a new lexical search every single time it encounters a strong/stressed syllable** in the incoming speech stream.

The key supporting experiment:

- **Method:** participants listened to a series of nonword stimuli, and had to press a button as fast as possible whenever they detected a real word embedded within a nonword.
- **Materials: a matched pair:**
  - Stimulus 1: ***mintave*** (/mɪnteɪv/) : this item has **two strong syllables**, and the second strong syllable begins right at the /t/.

    - Stimulus 2: ***mintesh*** (/mɪntəʃ/) : this item has **one strong syllable followed by one weak syllable** (the final syllable has the reduced schwa vowel, not a full stressed vowel).

- **Result:** participants were **slower** to detect the embedded real word 'mint' in *mintave* than in *mintesh*.
- **Interpretation:** because *mintave* contains a second strong syllable (at /t/), the MSS predicts the system automatically triggers a **second, brand-new lexical search** right at that point, effectively segmenting the input as 'min-tave.' This extra, competing search process makes it harder/slower to notice that 'mint' was embedded at the start. In *mintesh*, since the second syllable is weak (unstressed), no second search is triggered, so 'mint' is easier to detect.

Cross-linguistic generalization of MSS:

- **Dutch** - has a broadly similar rhythmic structure to English (alternating stressed/unstressed syllables), though without English's sharp distinction between full and reduced vowels (Quené & Koster, 1998; van Zon & de Gelder, 1993).
- **French** - has comparatively simple syllable structures and a rhythm based more on the **syllable** as the timing unit rather than on stress (i.e., 'syllable-timed' rather than 'stress-timed') (Cutler, Mehler, Norris & Segui, 1986; Mehler, Dommergues, Frauenfelder & Segui, 1981).
- **Japanese** - the relevant rhythmic unit is the **mora**, a unit of syllable *weight*: a long syllable (e.g., one with a long vowel) counts as **two** moras. Worked example: **Tōkyō** is analyzed as containing four moras: **to–o–kyo–o** (Otake, Hatano, Cutler & Mehler, 1993).
- There is also a body of developmental research examining **how infants acquire** the language-appropriate segmentation strategy for whichever language(s) they're being raised in.

The Possible Word Constraint (PWC)

- **Refinement to MSS:** it's been proposed that the MSS doesn't operate alone — it works together with the **Possible Word Constraint**, which is also claimed to hold across different languages (Norris, McQueen, Cutler, Butterfield & Kearns, 2001).
- **What the PWC does:** it ensures that when the speech input is segmented into words, the process is **exhaustive,** meaning no 'leftover' bits of sound are allowed to remain that couldn't possibly be a word on their own.

Summing Up the Unit-Size Trade-off

- Across the models surveyed, the proposed size of the pre-lexical unit ranges all the way from **sub-phonemic units** (phonetic features), through the **phoneme**, up to the **entire syllable**.
- **Larger units** (e.g., syllables) → the lexical search can only be initiated after a longer delay (you have to wait for the whole syllable to be identified before you can search), **but** the search produces a **smaller, more manageable set** of candidate words (on average, fewer words share a particular starting syllable than share a particular starting phoneme).
- **Smaller units** (e.g., phonemes) → the lexical search can begin much sooner, **but** it will initially produce a much **larger set** of candidate words to sort through, since many more words share any given starting phoneme.